

Week 10, Preimages

$$f: A \rightarrow B$$

$$a \mapsto f(a)$$

element

set

$$w \subseteq A$$

$$f(w) = \{ f(a) : a \in w \}$$

$$f(w) \subseteq B$$

$$\forall a \quad f(a) \in B \quad \forall a \in A$$

Useful: $\forall a \in w, f(a) \in f(w)$

$$P(A) \rightarrow P(B)$$

$$w \mapsto f(w)$$

"Preimage" or "inverse image"

Defn: Let $A \xrightarrow{f} B$ be a fn. Let $w \in B$.

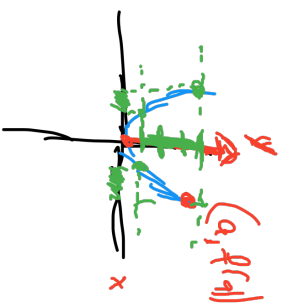
We define the preimage of w under f to be

$$f^{-1}(w) = \{a \in A \text{ s.t. } f(a) \in w\}$$

Caution!

THIS IS NOT RELATED TO THE INVERSE FUNCTION

Example, $f: \mathbb{R} \rightarrow \mathbb{R}$
 $x \mapsto x^2$



$$W = \mathbb{R}_{\geq 0} \quad f^{-1}(w) = \{a \in \mathbb{R} \text{ s.t. } f(a) \in \mathbb{R}_{\geq 0}\}$$

"a"

$$= \mathbb{R}$$

$$W = [1, 4]$$

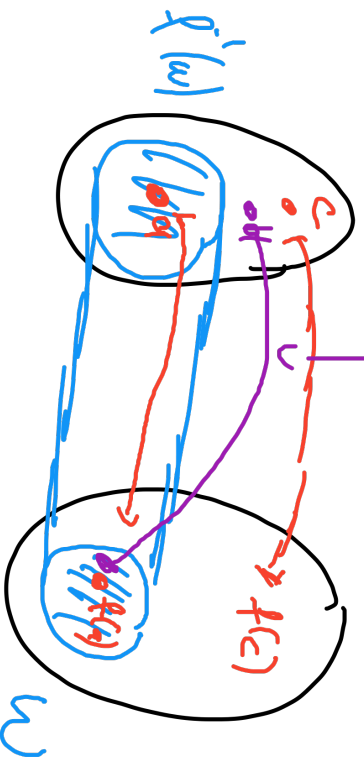
$$f^{-1}(w) = [-2, 2] \cup [-2, -1]$$

Use def:

$$a \in f^{-1}(w) \iff f(a) \in w$$

→ This is either what we know, or what we want to show

invalid picture



$$c \notin f^{-1}(w) \iff f(c) \notin w$$

$$A \xrightarrow{f} B \quad w = B \quad ?$$

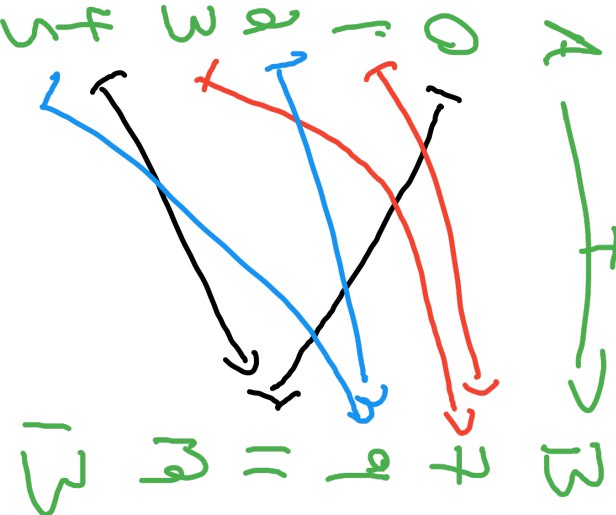
Comment:

$$f'(B) = \left\{ a \in A \text{ s.t. } \boxed{f(a) \in B} \right\} = A$$

Always true

(By the defn of f_{x_1})
 Codomain

Examples:



$$f(0) = f(4) = 11$$

$$f(1) = f(3) = 9$$

$$f(2) = f(5) = 13$$

$$f^{-1}(\{9, 11, 13\}) = \{a \in A \text{ s.t. } f(a) \in \{9, 11, 13\}\} \\ = \{0, 4, 2, 5\}$$

$$f^{-1}(\{7, 12, 3\}) = \{1, 3\}$$

$$f^{-1}(\{7, 3\}) = \{1, 3\}$$

$$f^{-1}(\{12, 13, 3\}) = \emptyset$$

Example: $A \xrightarrow{f} B$

$$W = \emptyset \subseteq B$$

$$f^{-1}(\emptyset) = \{ a \in A \text{ s.t.}$$

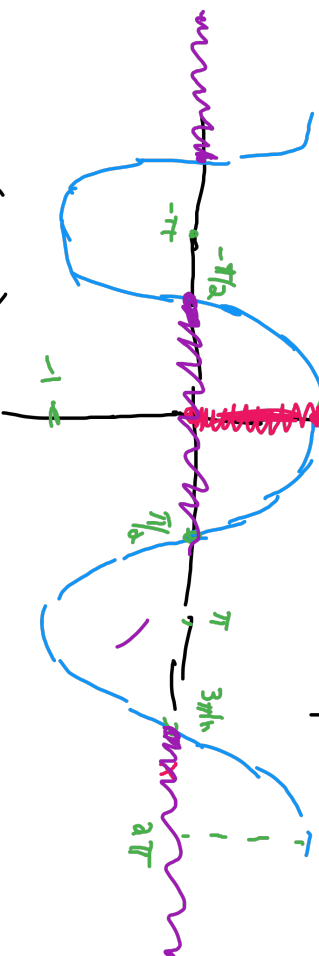
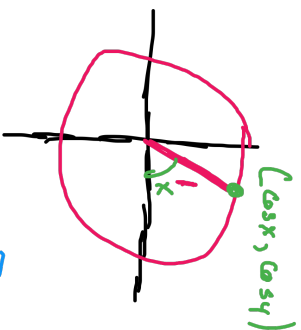
$$f(a) \in \emptyset \}$$

Always false

$$= \emptyset$$

Example: $f: \mathbb{R} \xrightarrow{\cos} \mathbb{R}$

$x \mapsto \cos x$



$f^{-1}([0,1]) = \{a \in \mathbb{R} \text{ s.t. } \cos a \in [0,1]\}$

$= \dots [-5\pi/6, -\pi/6] \cup [-\pi/2, \pi/2] \cup [3\pi/2, 5\pi/2] \cup [7\pi/6, 9\pi/6] \dots$

$(\mathbb{Z}) \bigcup_{n=-\infty}^{\infty} [-\pi/2 + 2n\pi, \pi/2 + 2n\pi]$

$f^{-1}(\mathbb{R}_{\geq 0}) \supseteq f^{-1}([0,1])$

$f^{-1}([-1,1]) = \mathbb{R}$

$f^{-1}([2,3]) = \emptyset$

$f^{-1}([-0.23, 0.17]) = \text{ bunch of intervals }$

(w/ no nice formula)



Solve $\cos a = 0.17$
 $a = \cos^{-1} 0.17$

$$\mathbb{R} \xrightarrow{f} \mathbb{R} \quad f^{-1}([1,4]) = [1,2] \cup [-2,-1]$$

$$x \mapsto x^2$$

$$[a,b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$

Proof: " $\supseteq$ " Let $x \in [1,2] \cup [-2,-1]$. Then $x \in [1,2]$ or $x \in [-2,-1]$

Then $1 \leq x \leq 2$ or $-2 \leq x \leq -1$.

$$\text{(wts: } x \in f^{-1}([1,4]) \text{ i.e., } f(x) \in [1,4] \text{ i.e., } 1 \leq x^2 \leq 4)$$

Case 1: $1 \leq x \leq 2$. Then squaring gives $1 \leq x^2 \leq 4$. Thus $x^2 = f(x) \in [1,4]$,
 thus $x \in f^{-1}([1,4])$.

Case 2: $-2 \leq x \leq -1$. Then squaring gives $1 \leq x^2 \leq 4$. Thus $x^2 = f(x) \in [1,4]$.

Thus $x \in f^{-1}([1,4])$.

" $\subseteq$ " Let $x \in f^{-1}([1,4])$. Then $f(x) = x^2 \in [1,4]$. Then $1 \leq x^2 \leq 4$.

Then $1 \leq x \leq 2$ or $-2 \leq x \leq -1$. Thus $x \in [1,2]$ or $x \in [-2,-1]$

Thus $x \in [1,2] \cup [-2,-1]$ □

18. For the following functions, compute the *inverse image* of the given subsets of the codomain. (No proofs are necessary.)

- (a) $f: \mathbf{Z} \rightarrow \mathbf{Z}, f(n) = 3n + 1$; $W_1 = \mathbf{E}$, the set of even integers, $W_2 = \{4\}$, $W_3 = \{1, 5, 8\}$;
 (b) $f: \mathbf{R} \rightarrow \mathbf{R}, f(x) = 3x + 1$; $W_1 = \{4\}$, $W_2 = \{1, 5, 8\}$, $W_3 = (4, \infty)$, $W_4 = (2, 4)$, $W_5 = \mathbf{Z}$, the set of even integers;
 (c) $f: \mathbf{R} \rightarrow \mathbf{R}, f(x) = \cos x$; $W_1 = [-1, 1]$, $W_2 = \{x \in \mathbf{R} \mid x \geq 0\}$, $W_3 = \mathbf{Z}$,
 (d) $f: \mathbf{R} \rightarrow \mathbf{R}, f(x) = e^x$; $W_1 = [-1, 0]$, $W_2 = \{x \in \mathbf{R} \mid x \geq 0\}$, $W_3 = \{1\}$,
 (e) $f: \mathbf{Z} \rightarrow \mathbf{Z}, f(n) = \begin{cases} n, & \text{if } n \text{ is even} \\ n-1, & \text{if } n \text{ is odd} \end{cases}$; $W_1 = \mathbf{E}$, $W_2 = \{1\}$, $W_3 = \{6\}$, $W_4 = \mathbf{O}$, the set of odd integers.

$$(a) f^{-1}(\mathbf{E}) = \mathbf{O}$$

" $\supseteq$ " Let $a \in \mathbf{O}$. (wts $a \in f^{-1}(\mathbf{E})$. i.e., $f(a) \in \mathbf{E}$)

Then $f(a) = 3a + 1$. Since a is odd, $3a$ is odd,

so $3a + 1$ is even. Thus $f(a) \in \mathbf{E}$, thus $a \in f^{-1}(\mathbf{E})$.

" $\subseteq$ " Let $a \in f^{-1}(\mathbf{E})$. Then $f(a) \in \mathbf{E}$. Thus $3a + 1$ is even.

Thus $3a$ is odd, so $a \in \mathbf{O}$. $\square$

$$e) f(n) = \begin{cases} n & n \in \mathbf{E} \\ n-1 & n \in \mathbf{O} \end{cases}$$

$$f(a) = 0 \quad f(a) = 2$$

$$f(1) = 0 \quad f(3) = 2$$

$$f^{-1}(\{1, 3\}) = \emptyset \quad f^{-1}(\{5, 6, 7\}) = \{6, 7\}$$

$$f^{-1}(\mathbf{E}) = \mathbf{Z}$$

Abstract proofs: $f: A \rightarrow B$
 $X, Y \subseteq B$

$$f^{-1}(X \cup Y) \subseteq f^{-1}(X) \cup f^{-1}(Y)$$

Proof: Let $a \in f^{-1}(X \cup Y)$. Then $f(a) \in X \cup Y$.

Then $f(a) \in X$ or $f(a) \in Y$.

(WTS: $a \in f^{-1}(X) \cup f^{-1}(Y)$. If $a \in f^{-1}(X)$ or $a \in f^{-1}(Y)$)

If $f(a) \in X$ or $f(a) \in Y$

Thus $a \in f^{-1}(X)$ or $a \in f^{-1}(Y)$

Thus $a \in f^{-1}(X) \cup f^{-1}(Y)$.

" $\supset$ " every step is reversible! i.e., an "if and only if" ($\Leftrightarrow$)

IE do the same proof backwards.

$$f^{-1}(X \cap Y) = f^{-1}(X) \cap f^{-1}(Y)$$

Switch "or" with "and" in prev. proof

"Let $a \in f^{-1}(X \cap Y)$ "

then $f(a) \in X \cap Y$.

Then $f(a) \in X$ and

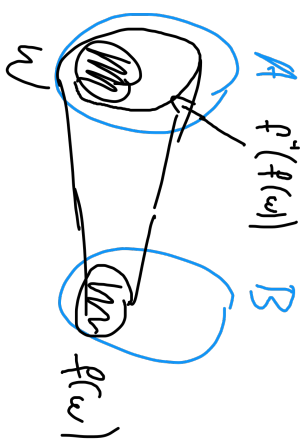
$f(a) \in Y$.

Then $a \in f^{-1}(X)$ and $a \in f^{-1}(Y)$.

Then $a \in f^{-1}(X) \cap f^{-1}(Y)$.

$$A \xrightarrow{f} B$$

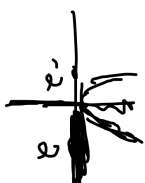
$$w \in A$$



$$\mathbb{R}^x \mapsto \mathbb{R}^x$$

$$f^{-1}(f(w)) = f^{-1}(\mathbb{R}_{\geq 0}) = \mathbb{R}$$

$$w = \mathbb{R}_{\geq 0} \quad f(w) = \mathbb{R}_{\geq 0}$$



$$\text{Claim } w \subseteq f^{-1}(f(w))$$

$$\text{Let } a \in w. \text{ (uns } a \in f^{-1}(f(w)) \text{) } \text{ I.E. } f(a) \in f(w)$$

$$\text{Then } f(a) \in f(w) \text{ by defn of image.}$$

$$\text{Then } a \in f^{-1}(f(w)), \quad \square$$

$$\text{(defn of pre-image.)}$$

$$\left. \begin{array}{l} a \in w \\ \Downarrow \\ f(a) \in f(w) \end{array} \right\} \text{defn of } f(w)$$

$$f^{-1}(f(w)) \subseteq w \text{ is false}$$

$$\mathbb{R} \subseteq \mathbb{R}_{\geq 0}$$

$$\text{for } w = \mathbb{R}_{\geq 0} \quad f(x) = \max(x, 0)$$